

Test Specification - 01: General Modules (Web and Mobile)							
Designed by: ARTICTON Development Team				Module: Shared Authentication, Accounts, Content, Progress, Security, and Support			
Objectives: Verify workflows and authoritative data shared by the ARTICTON web and mobile systems.				Test Platform: Google Chrome / Microsoft Edge and Android / iOS devices or emulators			
Test Case #	Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail	Performed by / Date
G-01	Authenticate shared user accounts	1. Attempt login on web and mobile using the same valid account. 2. Repeat with an incorrect password and an unknown account.	Registered account Correct password Incorrect password	Both clients accept the valid credentials and reject invalid credentials without creating a session. Error messages remain clear and consistent.			
G-02	Apply role-based access across both clients	1. Sign in as student, faculty/staff, and administrator on each supported client. 2. Review the landing view and available actions.	Student, faculty/staff, and administrator accounts	Each role reaches only its authorized dashboard and tools. Restricted actions are hidden or denied on both platforms.			
G-03	Restore and terminate sessions safely	1. Sign in on each client. 2. Restart/refresh the client and verify session restoration. 3. Sign out and attempt to reopen a protected view.	Any valid account	An active valid session is restored. Signing out clears access, returns to authentication, and prevents protected data from reopening.			
G-04	Synchronize profile information	1. Update an allowed profile field. 2. Save the change. 3. Open the same account on the other client.	Valid first name, last name, and supported profile values	The saved profile belongs only to the current user and the same authoritative value appears on web and mobile.			
G-05	Synchronize assessment and progress records	1. Complete a supported assessment on one client. 2. Open the progress/dashboard view on the other client. 3. Compare score, status, and completion.	Shared student account Supported module assessment	The server stores one authoritative result and both clients show matching score, pass/fail state, and progress without duplicate credit.			
G-06	Publish approved learning content to both clients	1. Submit a module or question change as faculty. 2. Approve it as administrator. 3. Open the affected content on web and mobile.	Valid change request and administrator approval	Only approved content is published, and both clients load the same current revision while rejected/stale requests do not publish.			
G-07	Protect private and administrative data	1. Attempt to read or modify another user's private data as a student. 2. Attempt an admin-only operation. 3. Repeat the intended operation with an authorized role.	Student and administrator accounts	Unauthorized reads/writes are denied on both platforms; authorized operations succeed and expose only the fields required for the workflow.			
G-08	Submit and manage shared support requests	1. Submit a valid support request from an available student client. 2. Open the admin support queue on web. 3. Change its status.	Name, email, subject, message, optional screenshot	One ticket is created through the approved workflow, appears in the admin queue, and moves from Open to In progress to Resolved.			

Test Specification - 02: Web Application Test Cases

Designed by: ARTICTON Development Team				Module: ARTICTON Web Learning and 3D Practical Assessment System			
Objectives: Verify web-specific learning, 3D simulation, practical scoring, administration, and faculty workflows.				Test Platform: Google Chrome and Microsoft Edge (current desktop versions)			
Test Case #	Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail	Performed by / Date
W-01	Log in with valid and invalid credentials	1. Open ARTICTON. 2. Enter a registered student email and correct password. 3. Select Login. 4. Log out, then repeat with an incorrect password.	Registered student account Correct password Incorrect password	Valid credentials open the student dashboard. Invalid credentials do not create a session and display a clear error message.			
W-02	Route users according to role	1. Log in separately as student, faculty/staff, and administrator. 2. Observe the first page and available navigation for each account.	One active account for each supported role	Student opens the learning dashboard; faculty/staff opens faculty tools; administrator opens the admin dashboard. Unauthorized tools are not shown.			
W-03	Sign out and protect the session	1. Log in. 2. Select Logout. 3. Attempt to return to the previous protected view using browser navigation.	Any valid user account	The session ends, the landing/login view appears, and protected data cannot be used without signing in again.			
W-04	Open Module 1 and record learning progress	1. From the student dashboard, open Module 1. 2. Review hardware content and complete available parts. 3. Return to the dashboard and reopen the module.	Student account with Module 1 access	Module 1 loads correctly, completed content is saved, dashboard progress updates, and the student can resume the latest visit.			
W-05	Use guided assembly for AMD and Intel	1. Open Module 2. 2. Select AMD and follow the ordered assembly steps. 3. Repeat with Intel. 4. Try advancing with a prerequisite incomplete.	Student account AMD and Intel platform selections	The correct 3D models and platform-specific sequence appear. Completed steps persist, and prerequisite rules prevent an invalid sequence.			
W-06	Use guided disassembly for AMD and Intel	1. Open Module 3. 2. Select each platform in turn. 3. Follow the removal sequence and attempt one out-of-order action.	Student account AMD and Intel platform selections	The correct removal sequence is shown, valid actions advance progress, and an out-of-order action gives feedback without corrupting saved progress.			
W-07	Complete a module quiz and store the score	1. Open an available module quiz. 2. Answer all questions. 3. Submit the quiz. 4. Refresh the dashboard.	Student account Known correct and incorrect answers	The quiz is scored once, the result and completion state are saved, and the dashboard/admin summary shows the same score.			
W-08	Complete an assembly practical in the correct order	1. Open an AMD or Intel full assembly practical. 2. Place all components in the required sequence. 3. Finish the assessment.	Student account Correct shared assembly sequence	The checklist advances, completion reaches 100%, deductions are zero, and a score of at least 75% is marked Passed and saved.			
W-09	Score sequence errors in an assembly practical	1. Start a full assembly practical. 2. Select a component before its prerequisite. 3. Continue and finish the assessment.	Student account At least one intentional order error	The system records an order mistake, applies the configured capped deduction, shows feedback, and determines Passed/Failed from the final unrounded score.			

W-10	Complete a disassembly practical	1. Open an AMD or Intel full disassembly practical. 2. Remove parts in the required order. 3. Finish and return to the dashboard.	Student account Correct shared disassembly sequence	The correct sequence is enforced, the final score and elapsed time are stored, and the completed practical appears on the dashboard.			
W-11	Display student progress to faculty/admin	1. Sign in as faculty or administrator. 2. Open the student monitoring area. 3. Search/select a student with quiz and practical records.	Faculty/admin account Student with saved progress	The authorized user can view the student's supported module, quiz, and practical summaries; private profile fields are not exposed unnecessarily.			
W-12	Submit and review a module content change	1. Sign in as faculty. 2. Edit supported module content and submit a summary. 3. Sign in as administrator. 4. Approve or reject the request.	Faculty account Administrator account Valid change summary	The request enters the admin approval queue. Approval updates published content; rejection preserves approved content and records the decision.			
W-13	Manage question-bank content with role restrictions	1. Sign in as faculty/staff and create or edit a supported question. 2. Confirm the saved item. 3. Attempt the same operation as a student.	Faculty/staff account Student account Valid question and choices	Authorized staff can save valid question data. Student write attempts are denied and do not modify the question bank.			
W-14	Submit and process a support ticket	1. As a student, submit a valid support request. 2. As an administrator, locate it by search/filter. 3. Change status from Open to In progress to Resolved.	Name, email, subject, message Optional screenshot	The ticket is created through the approved workflow, visible to the administrator, searchable, and its status updates without altering protected ticket content.			
W-15	Save user settings and profile changes	1. Open Settings/Profile. 2. Change allowed preferences and profile fields. 3. Save, refresh, and sign in again.	Valid display/profile values and user preferences	Allowed values persist across refresh and sign-in, invalid values are rejected, and changes affect only the current user.			
W-16	Use the procedure assistant safely	1. Open a supported module or practical. 2. Ask a relevant procedural question. 3. Ask an unrelated or unsafe question. 4. Test voice-over when available.	Relevant hardware question Unrelated/unsafe prompt	The assistant gives contextual learning guidance, avoids enabling unsafe actions or revealing answers improperly, handles fallback mode, and voice-over reads the response when enabled.			

Test Specification - 03: Mobile Application Test Cases							
Designed by: ARTICTON Development Team				Module: ARTICTON Flutter Mobile Learning and Assessment Application			
Objectives: Verify mobile navigation, four learning modules, server-backed assessments, progress, practice exams, and role tools.				Test Platform: Android and iOS phone/tablet devices or emulators			
Test Case #	Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail	Performed by / Date
M-01	Restore a mobile session from the splash screen	1. Sign in and close the application. 2. Reopen the application with the account still authenticated. 3. Repeat after signing out.	Android/iOS device or emulator Valid student account	The splash screen restores a valid server-backed profile and opens the correct account scope. After sign-out, it opens the login flow.			
M-02	Use responsive mobile navigation	1. Open the student dashboard in portrait. 2. Navigate through Dashboard, Modules, Exams, and Profile. 3. Rotate to landscape or use a wide device.	Phone and tablet/wide emulator	Portrait uses the bottom navigation bar; wide/landscape uses the navigation rail. The selected page and readable layout remain stable.			
M-03	Open all four learning modules	1. Open Modules. 2. Visit Module 1: PC Parts, Module 2: Disassembly, Module 3: Assembly, and Module 4: Software & Networking. 3. Review approved lesson cards.	Authenticated student Published or starter content	Each module displays its correct title and content. Approved content is preferred, and the app remains usable when starter content is used.			
M-04	Start a supported pre-test or post-test	1. Open a module pre-test or post-test. 2. Start the assessment. 3. Verify question text and two to four options. 4. Attempt an unsupported assessment ID.	One of the eight module assessment IDs Unsupported ID	Supported assessments create a server attempt with valid questions. Unsupported IDs are rejected before a callable request is made.			
M-05	Submit assessment answers and calculate results	1. Answer every question. 2. Submit the attempt. 3. Compare correct count, total, percentage, and pass state. 4. Try an answer index outside 0-3.	Valid attempt ID Valid answer map Invalid answer index	The server result is shown and clamped to 0-100. Invalid answer indexes are rejected. A module post-test reaches Passed at 70% or higher.			
M-06	Complete guided module content	1. Finish the guided content for a module. 2. Trigger completion. 3. Return to the dashboard and reopen progress.	Authenticated owner account Supported module ID	Completion is sent through the protected completeGuidedModule callable and the dashboard reflects the saved content component for that user.			
M-07	Calculate module and overall progress	1. Complete a pre-test, module content, and post-test. 2. Open Dashboard and Profile. 3. Compare displayed module and overall percentages.	Student with saved module_scores records	Progress uses 20% pre-test, 50% content completion, and 30% post-test percentage. The overall value is the rounded average of four modules.			
M-08	Open mobile practice exams under the configured gate	1. Open Exams below and above 70% overall progress. 2. Test Practice Exam 1 and Practice Exam 2. 3. Repeat with the development bypass setting noted.	Student progress below/above 70% Development and production configurations	Production configuration locks exams below 70% and unlocks them at 70%. The current development bypass permits testing and is clearly treated as non-production behavior.			

M-09	Edit profile, submit support, and sign out	1. Open Profile. 2. Save valid profile changes. 3. Submit a support request. 4. Sign out.	Valid profile fields Support subject and message	Profile changes save for the current account, the support request is accepted, and sign-out clears navigation history and returns to login.			
M-10	Switch theme and preserve readable UI	1. Switch between light and dark modes. 2. Navigate across Dashboard, Modules, Exams, and Profile. 3. Restart the app.	Phone and tablet/wide emulator	Theme selection applies consistently, text and controls remain readable, and the selected mode persists according to the theme controller.			
M-11	Use faculty mobile content and question editors	1. Sign in as faculty/staff. 2. Draft a supported module change. 3. Edit valid question content. 4. Submit both for approval.	Faculty/staff account Valid cards, summary, questions, options, answers, and explanations	Valid requests are saved for admin review; incomplete or invalid assessment data is rejected and does not publish directly.			
M-12	Review mobile approval queues as administrator	1. Sign in as administrator. 2. Open module and question approvals. 3. Approve one current request and reject another.	Administrator account Pending current requests	Approved content becomes the published revision, rejected content remains unpublished, and stale/already-reviewed requests cannot overwrite newer data.			

Execution note: Complete the final three columns during the formal test run. Record evidence or defect IDs in Actual Results when a case fails.